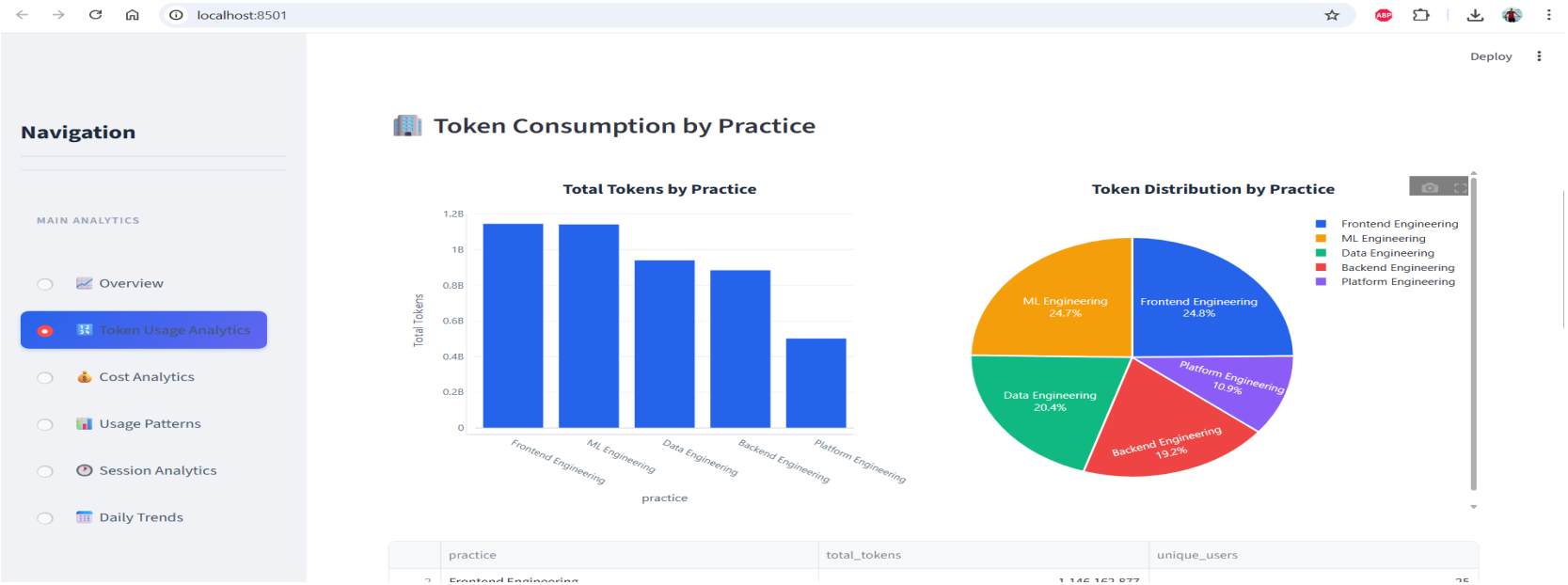
Claude Code Usage Analytics Platform

Project Overview

Goal: Build an analytics platform that processes Claude Code telemetry data and transforms raw event streams into actionable insights about developer behavior.

Dataset:
- Telemetry events (token usage, code generation requests)
- Session metadata
- User roles and project types

Tech Stack (example):
- Python
- Pandas – data processing
- SQL database – structured storage
- Streamlit – interactive dashboard
- FastAPI – API access to analytics data



System Architecture

Data Pipeline Overview:

Raw Telemetry Data (JSON / CSV)
↓
Data Ingestion & Cleaning (Python + Pandas)
↓
Structured Storage (SQL Database)
↓
Analytics Layer
↓
Interactive Dashboard (Streamlit)
↓
API Endpoints (FastAPI)

The architecture separates ingestion, storage, analytics, and visualization layers to ensure scalability and maintainability.

Key Insights from Telemetry Data

Example Insights:

Token Usage by Role
• Developers generate significantly more tokens compared to other roles.

Peak Usage Times
• Highest activity appears during afternoon working hours.

Code Generation Patterns
• Code generation events dominate the telemetry dataset.

Session Behavior
• Longer sessions typically correlate with higher token consumption.

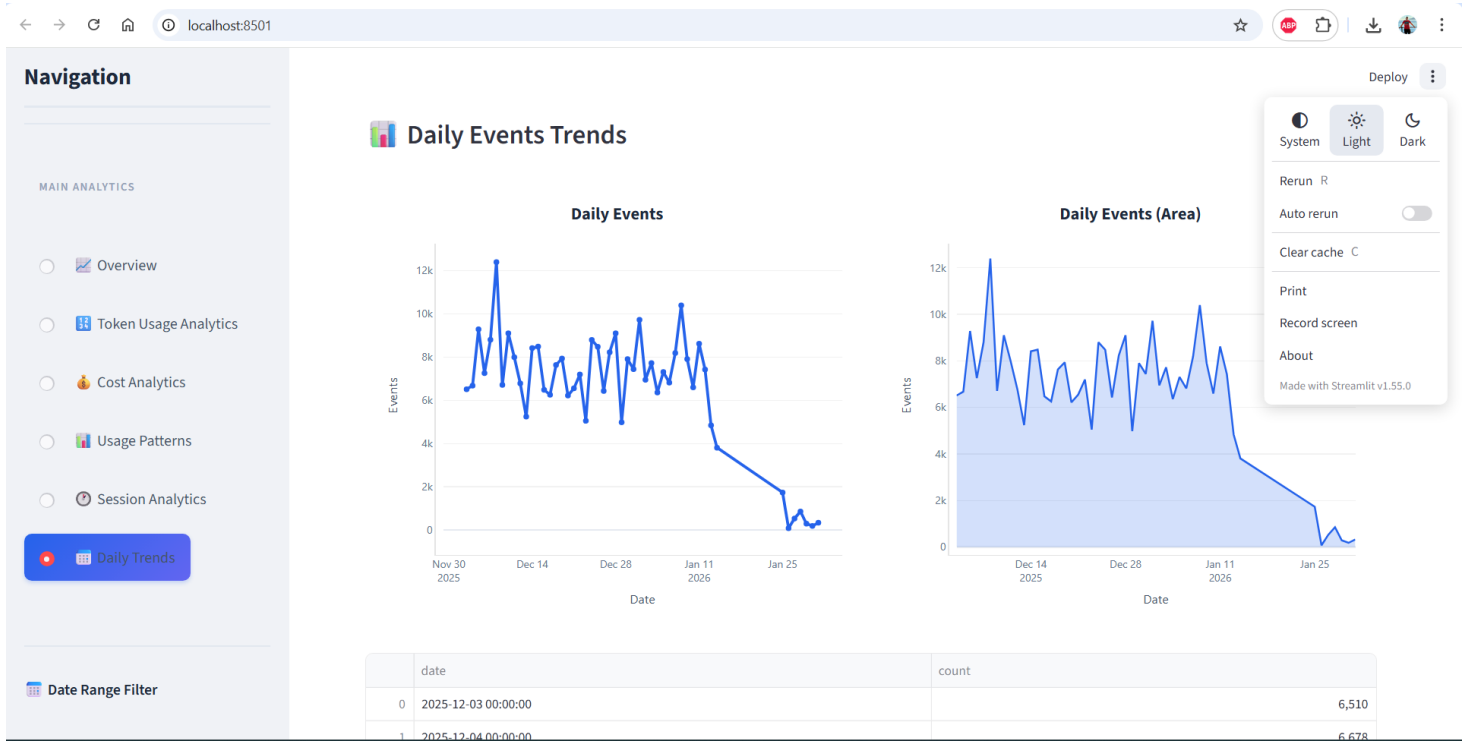
These insights help understand how developers interact with Claude Code in real workflows.

Interactive Dashboard

Dashboard Features:

- Interactive filtering by user role and project type
- Visualization of token usage trends
- Session activity monitoring
- Code generation frequency analytics

The dashboard allows stakeholders to quickly explore developer behavior patterns and system usage metrics.



Conclusion & Future Improvements

Conclusion:

The platform successfully transforms raw Claude Code telemetry data into structured analytics and visual insights. By combining automated data processing with interactive visualization, the system enables better understanding of developer behavior and AI-assisted coding patterns.

Future Improvements:

- Real-time telemetry streaming
- Anomaly detection for unusual activity